

# Scott Ensel

Ph.D. - Postdoctoral Researcher

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Pittsburgh, PA 

Biomedical engineering PhD with 8+ years of experience in medical devices, sensorimotor control, neuromodulation, fMRI, and machine learning based neural decoding across human and non-human primate research. Leads first-in-human spinal cord stimulation clinical trials to restore movement after neurodegenerative diseases and stroke and contributed to multiple high-impact publications. Proven ability to collect and analyze behavioral and electrophysiological data, interface with clinical surgeons, researchers, and collaborate cross-functionally. An experienced scientific communicator with multiple conference presentations. Driven to translate complex neuroscience into practical, user-centered technologies that improve quality of life.

## EDUCATION

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|--------------------------------------------------------------------------------------------------------|------------------|
| University of Pittsburgh                                                                               | Pittsburgh, PA   |
| <b>Ph.D. Student – Bioengineering (Neural Engineering)</b>                                             | 2020 - 2025      |
| University of Michigan                                                                                 | Ann Arbor, MI    |
| <b>M.S in Electrical &amp; Computer Engineering (Signal and Image Processing and Machine Learning)</b> | 2017 - 2019      |
| <b>M.S in Biomedical Engineering (Bioelectric and Neural Engineering)</b>                              | 2017 - 2019      |
| University of Maryland                                                                                 | College Park, MD |
| <b>B.S in Bioengineering</b>                                                                           | 2013 - 2017      |

## RESEARCH EXPERIENCE

**Rehabilitation and Neural Engineering Laboratory** Pittsburgh, PA  
*Postdoctoral Researcher* October 2025 - Present

**Project: “Targeted deep brain stimulation of the motor thalamus improves speech and swallowing motor functions after cerebral lesions”**

- Piloted execution of a clinical trial using deep brain stimulation in three stroke participants to restore speech, swallowing, and movement
- Authored FDA IDE submission for first-in-human deep brain stimulation trial investigating dysphagia restoration following traumatic brain injury

**Rehabilitation and Neural Engineering Laboratory** Pittsburgh, PA  
*Doctoral Researcher* September 2020 - September 2025

**Thesis: “Decoding Spinal Cord Function: Capturing Altered Motor Functional Connectivity Through fMRI Biomarkers in Neurological Disorders”**

- Piloted design and execution of first in-human study using spinal cord stimulation to reverse maladaptive motor and neurodegenerative effects in six spinal muscular atrophy (SMA) participants, quantified through clinical exams, kinematics, electrophysiological, and neuroimaging analyses
- Lead and designed imaging data collection pipeline for a pilot clinical trial of spinal cord stimulation in eight participants with stroke to regain motor function of upper limbs using spinal cord stimulation
- Created a novel cortico-spinal fMRI sequence for single subject assessment to sense and characterize novel biomarkers of functional brain changes in neurological disorders

**Cortical Neural Prosthetics Laboratory** Ann Arbor, MI  
*Research Assistant* September 2018 - July 2020

- Implemented a real-time convolutional neural network in PyTorch to decode continuous individual finger positions from non-human primate neural spiking data that achieved a 36% increase in throughput over ReFIT Kalman Filter
- Implemented a real-time neural network in PyTorch to classify hand movement and then decode individual finger position from regenerative peripheral nerve interfaces in upper-limb amputees to control myoelectric prosthetic
- Designed a computer network and built computer hardware that integrated into current data collection pipeline to optimize neural network decoder development

**Children’s National Health System - Sheikh Zayed Institute** Washington D.C.  
*Research Intern* June 2016 - June 2017

- Lead collection and segmentation of radiology data in preparation for multiple research projects that applied machine learning to achieve pediatric surgical innovation

## WORK EXPERIENCE

**Terumo Heart, Inc – Optical Engineering Team** Ann Arbor, Michigan  
*Optical Engineering Intern* May 2018 - January 2019

- Created a functional pipeline integrating advanced image segmentation techniques to automate testing of

## RELEVANT SKILLS

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BCI | SCS | DBS | Neuromodulation | Medical Devices | Electrophysiology | Electromyography | Biosensors | Kinematic & Neural Analysis | Non-human primate training and neural recordings | Clinical human-subject research (Stroke, SMA, SCI) | Experimental Design | Cross-disciplinary collaboration | User-Centered Research | IDE Submission | Adobe Illustrator | Microsoft Office (Word, Excel) | Technical Writing | Grant Writing | Oral presentations

## TECHNICAL SKILLS

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Python (PyTorch, scikit-learn) | MATLAB | Bash Scripting | Julia | R | Machine Learning | Statistics | Signal Processing | Image Processing | fMRI | FSL | Spinal Cord Toolbox

## JOURNAL ARTICLES

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### **Nature Medicine - 2026**

de Freitas, R.M., Bhatia, S., Sorensen, E. *et al.* Spinal cord stimulation for upper limb motor function in people with chronic post-stroke hemiparesis: a feasibility trial. *Nat Med* (2026).

### **Brain - 2025**

Simon, C. *et al.* Proprioceptive synaptic dysfunction is a key feature in mice and humans with spinal muscular atrophy, *Brain*, 2025, <https://doi.org/10.1093/brain/awaf074>

### **Nature Medicine - 2025**

Prat-Ortega, G., Ensel, S., Donadio, S. *et al.* "First-in-human study of epidural spinal cord stimulation in individuals with spinal muscular atrophy". *Nat Med* (2025). <https://doi.org/10.1038/s41591-024-03484-8>

### **Neuron - 2025**

Balaguer, J. *et al.*, Neural mechanisms underlying the recovery of voluntary control of motoneurons after paralysis with spinal cord stimulation, *Neuron*, (2025)

### **Communications Biology- 2024**

Ensel, S., Uhrig, L., Ozkirlı, A. *et al.* Transient brain activity dynamics discriminate levels of consciousness during anesthesia. *Commun Biol* 7, 716 (2024).

### **Nature Medicine - 2023**

Powell, M., Verma, N., Sorensen, E., Carranza, E., Boos, A., Fields, D., Roy, S., Ensel, S., Barra, B., Balzer, J., Goldsmith, J., Friedlander, R., Wittenberg, G., Fisher, L., Krakauer, J., Gerszten, P., Pirondini, E., Weber, D. and Capogrosso, M. "Epidural stimulation of the cervical spinal cord for post-stroke upper-limb paresis. *Nature Medicine* 29, 689-699 (2023). <https://doi.org/10.1038/s41591-022-02202-6>

### **Nature Communications - 2022**

Willsey, M.S., Nason-Tomaszewski, S.R., Ensel, S.R. *et al.* Real-time brain-machine interface in non-human primates achieves high-velocity prosthetic finger movements using a shallow feedforward neural network decoder. *Nat Commun* 13, 6899 (2022).

### **IEEE Transactions on Medical Imaging - 2018**

A. R. Porras *et al.*, "Locally Affine Diffeomorphic Surface Registration and Its Application to Surgical Planning of Fronto-Orbital Advancement," in *IEEE Transactions on Medical Imaging*, vol. 37, no. 7, pp. 1690-1700, July 2018.

## CONFERENCE PRESENTATIONS

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### **Organization for Human Brain Mapping, Seoul, South Korea, June 2024**

Lumbar Spine fMRI to Quantify Efficacy of Spinal Cord Stimulation Therapy in Spinal Muscular Atrophy, Ensel, S., *et al*

### **Organization for Human Brain Mapping, Montreal, Canada, July 2023**

Signature of Consciousness in the Transient Dynamics of fMRI Brain Activity, Ensel, S., *et al*

### **University of Pittsburgh BIOE day Spotlight Presentation, Pittsburgh, PA, 2022**

Resting State Dynamics as a Cortical Signature of Consciousness, Ensel, S., *et al*

## HONORS AND AWARDS

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### **CMU-Pitt BRIDGE Center seed funding • 2025**

Awarded to conduct a collaborative research project at BRIDGE Center.

Decoding Spinal Cord Function: Advancing Clinical Impact through fMRI Biomarkers in Neurological Disorders

### **Big Idea Competition • 2025**

Awarded third place prize to support student driven innovation

Lexi Medical is developing a novel deep brain stimulation (DBS) system to restore swallowing to individuals with post-stroke muscle weakness or paralysis.

### **CMU-Pitt BRIDGE Center seed funding • 2024**

Awarded to conduct a collaborative research project at BRIDGE Center.

Decoding Spinal Cord Function: Advancing Clinical Impact through fMRI Biomarkers in Neurological Disorders

## **SCIENTIFIC SERVICE**

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Reviewed for **APL Bioengineering** 2025

Reviewed for **IEEE Transactions on Neural Systems and Rehabilitations Engineering** 2025

Reviewed for **Journal of Neural Engineering and Rehabilitation** 2024

## **INVITED RESEARCH VISITS**

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**Université Paris-Saclay (Neurospin) – October 2024**

## **PROFESSIONAL DEVELOPMENT**

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**2025 Neurotech Course Workshop**

## **TEACHING**

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**University of Pittsburgh – Spring 2022**

BIOENG 1580: Biomedical Applications of Signal Processing

**University of Pittsburgh – Fall 2021**

BIOENG 2167: Managing Medical Product Innovation

**University of Michigan – Fall 2018**

ASTRO 142: Big Bang, Fall 2018

**University of Maryland – Fall 2017**

BIOE 437 (Formerly BIOE 698V): Computer-Aided Design in Bioengineering

**University of Maryland – Fall 2015**

CHEM 132 General Chemistry I Laboratory